W01W02: Python Basic Data Types and Functions

Objective

This worksheet is designed to enhance your understanding of basic Python data types (string, integer and float) and functions. You will use commands and the built-in functions *type()*, and *print()*, calculate the radius and perimeter of a circle and finally create functions that calculate those values.

Part 1: Understanding Data Types

Create a file named *type.py* and write four lines of code. At each line of code, use the type() function to determine the data type of the following values.

E.g.,
print(type (42))

a. "Hello, World!"

b. 3.14159

c. 18

d. True

Part 2: Commenting your code

For each line of code, write a descriptive comment. As a reminder, lines that start by "#" are ignored by the compiler and they are not executed. If you want to comment multiple lines, use triple quotes (""") at the start and the end of each block.

Part 3: Using Variables

Python doesn't have constants (values that cannot change during the execution of a program. For readability and clarity, programmers define constants using uppercase letters. This is a coding convention, as it is not a strict rule, but it is be used within the Python community.

In this exercise, instead of calling the type() built-in function directly inside the print() function, write a variable for each item in Part 1 and then print it. For the (b) value 3.14159 use the name "PI", indicating that **π** is a constant.

E.g.,
number = 42
print (number)

It is a good practice to define our variables at the start of the program. For that reason, add your four variables at the top of your type.py file and call the print functions afterwards.

Part 4: Modifying Variables

Define a variable named "*radius"* that is equal to 2.5 and use PI to calculate its area and perimeter. Store the area inside a variable named *area (π * r * r)* and the perimeter (2 * *π* * *r)* in a variable named *perimeter*. Print the variable and perimeter. The result shall be the following:

```
area: 19.6349375

perimeter: 15.70795
```

Part 5: Functions

Define your constant *PI* at the start of your code. Then define two functions (this order will ensure that your functions are aware of *PI*:

1. The first function takes as input the radius and prints "area: " followed by the area of the circle.
2. The second function takes as input the radius and print "perimeter: " followed by the perimeter of the circle.
3. Call the functions giving them as input the variable *radius*, which is equal to 2.5.

The results shall be the following (unchanged):

```
area: 19.6349375

perimeter: 15.70795
```

Part 6: Returning values in functions

Modify your functions, so that they return the calculated value (area and perimeter respectively). Call the function, store the returned value in a variable and then print the results.

The result shall be the following (unchanged):

```
area: 19.6349375
perimeter: 15.70795
```